FAQ

Here are some questions we get a lot. If you have more specific questions, please ask in the forum.

Q. What is woodgas?

A. Woodgas is a flammable gas released by wood when it is heated. When you see flames in a fire, that is actually woodgas burning. If we capture the gas, we can burn it in an engine. Read more about the Basics of Gasification.

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Q. How do you run an engine on woodgas?

A. A spark ignition engine needs a flammable gas to run. Gasoline engines must first vaporize the liquid fuel. Woodgas is produced in gaseous state, so it can be piped straight to the engine, mixed with air and drawn in. Woodgas burns slowly, so the spark needs to be advanced quite a bit. Compression can be increased as well.

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Q. Doesn't it hurt the engine?

A. Sometimes you will hear that burning woodgas will harm your engine, or lead to shorter life. This is wrong. The grain of truth is that during WWII, gas filtration was very poor or non-existent, letting all sorts of ash, soot and even tar into the engines, causing premature wear or failure. With proper filters from modern materials, your engine may last even longer than before. See the Cooling & Filtration page for details.

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Q. Can I compress it at home and drive on compressed woodgas?

A. That would be nice! Unfortunately, woodgas is a extremely low BTU fuel - it only has 150 BTUs per cubic foot. Woodgas is used in a 1:1 fuel air mixture. Even if you compressed it to 3600psi the range of a 100 gallon tank would be less than two miles. And there's another problem. When you compress woodgas you are forcing the molecules closer together. The free hydrogen (H_2) will steal the O off of the CO, making H_2O and C (water and carbon). Instead of woodgas you have soot and water vapor. This will not power an engine.

Nature has already done an excellent job of compressing this bulky fuel - it's called wood!

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Q. Can you use woodgas to generate electricity?

A. Yes! This is a good source of power for off-grid life. Stationary woodgas production is more involved than mobile applications. In a vehicle, the bumps in the road help eliminate fuel bridging and keep the grate shaken to keep the ash moving. Stationary setups need to have a good shaker system in place. Moreover running woodgas derates the HP output of a gasoline engine by at least 25% - this is a problem when trying to get the rated KW out of a genhead.

Running a woodgas generator for an hour is fairly simple, but for continuous use it becomes less attractive. There are problems like refueling and ash removal, all while working on a very warm unit. Woodgas for electricity production is best suited to short (1 or 2 hour) runs to charge low batteries. Make your power in the morning and shut it down, rather than letting the generator idle along all day.

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Q. You are going to cut down all the trees! Shame on you!

A. That's not a question, but I will answer it anyway. Woodgas is usually made from waste wood. Chips, sawmill slabs, off-cuts, slash, and tree trimmings can all provide plenty of fuel without felling a tree. In fact, there is so much extra wood around that it is common practice to burn it off - simply to get it out of the way. Anyone in the country has seen burning brush piles.

While it might be true that there are not enough trees for everyone to run on woodgas, it is such a labor intensive fuel that only a few enthusiasts are using it. If gasoline got scarce, not many people would be willing or able to use woodgas to continue driving. There are far more people burning wood to heat houses than running woodgas. Yet deforestation is not imminent. Until woodgas goes mainstream, it is a moot point. Read more about deforestation. (/library/deforestation)

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Q. Can you use other fuels besides wood?

A. Yes, other carbon based fuels can be used. I know of at least one person who has run his unit on dried chicken dung! Jeff Davis has done a fair amount of testing with agglomerated biomass. One thing to watch out for is the ash content. High ash fuels like rice hulls will lead to slag forming on the grate. Slag is melted ash, and it can be tough to remove.

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Q. So does it spew lots of smoke?

A. There is no smoke visible from a running gasifier. Smoke is being produced and consumed, but none escapes to the world. During startup there is some smoke from the gasifier, which is not yet hot enough to make woodgas. Also at shutdown if the unit is not perfectly sealed, some wisps of smoke may escape. Occasionally a gasifier may "puff back", which releases a large burst of smoke. But this is very rare on a well made unit.

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Q. Where can I get plans for a gasifier?

A. Check out the Resources section, there are some plans available. Honestly, I don't recommend building any of them right off the shelf, except Wayne's plans. There are many parameters involved, and you need to have a solid understanding of gasification and the application you are building for. Read all the available materials, and then you will be equipped to design one that works for you.

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About Drive On Wood

A community of gasification enthusiasts hosted by Wayne Keith, resident expert & woodgas pioneer.

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